

AS Chemistry homework · 3.6 Intermolecular forces

CIE 9701 AS Chemistry · 18 marks · about 40 minutes

Topic: 3.6 Intermolecular forces, electronegativity and bond properties — hydrogen bonding (N–H, O–H), van der Waals' forces (id-id, pd-pd), dipole moments, and relative strength of bonding vs intermolecular forces. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; use syllabus terminology throughout.

Section	Content	Marks
A	Multiple choice	8
B1	2023 MJ Paper 22 Question 1(b)(iii)	1
B2	2023 MJ Paper 22 Question 1(b)(iv)	2
B3	2021 MJ Paper 22 Question 2(c)(iii)	2
B4	2025 MJ Paper 22 Question 1(d)(i)	3
B5	2021 ON Paper 22 Question 1(a)(ii)	2
Total		18

Section A — Multiple-choice questions (8 marks)**A1.**

[1]

- 5 In which substance are the only intermolecular forces temporary dipole-induced dipole attractions?

- A** hydrogen chloride
B methanol
C octane
D water

2021 FM Paper 12 Question 5

A2.

[1]

- 5 Descriptions of the bonding in three substances are given.

substance 1 strong covalent bonds between atoms, permanent dipole-dipole attractions between molecules

substance 2 strong covalent bonds between atoms, weak forces between molecules

substance 3 strong covalent bonds between atoms, hydrogen bonding between molecules

Which compounds could be substances 1, 2 and 3?

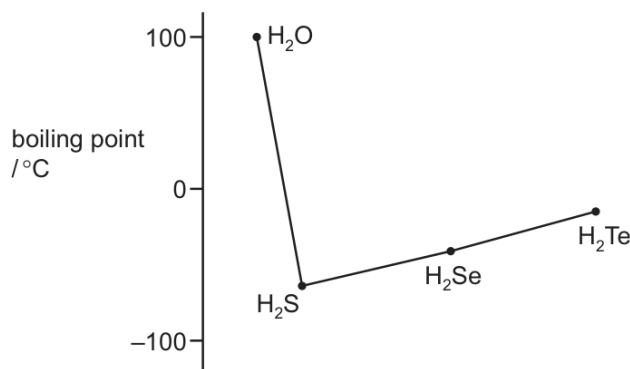
	substance 1	substance 2	substance 3
A	CH ₃ OH	Al ₂ Cl ₆	CH ₂ Cl ₂
B	Al ₂ Cl ₆	CH ₂ Cl ₂	CH ₄
C	CH ₂ Cl ₂	CH ₄	CH ₃ OH
D	CH ₄	CH ₃ OH	H ₂ O

2021 MJ Paper 13 Question 5

A3.

[1]

- 8 The graph shows the boiling points of the hydrogen compounds of Group 16 elements.



Which statement correctly explains why water does **not** fit the trend of the other compounds?

- A There are fewer electrons in the oxygen atoms so there is less shielding of the nuclear charge.
- B There are strong hydrogen bonds in water but **not** in the other compounds.
- C The covalent bonds in water are much stronger than in the other compounds.
- D The water molecules are smaller and so have stronger van der Waals' forces.

2024 MJ Paper 11 Question 8

A4.

[1]

- 8 Consider the following four compounds.

- 1 (CH₃)₃CH
- 2 CH₃CH₂CH₂OH
- 3 CH₃CH₂CH₂SH
- 4 CH₃CH₂CH₂CH₃

What is the order of increasing boiling point of the compounds (lowest first)?

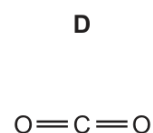
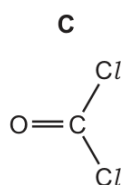
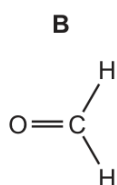
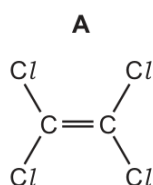
- A 1 → 4 → 2 → 3
- B 1 → 4 → 3 → 2
- C 4 → 1 → 2 → 3
- D 4 → 1 → 3 → 2

2024 MJ Paper 12 Question 8

A5.

[1]

- 6 Which molecule has the largest overall dipole?



2021 ON Paper 12 Question 6

A6.

[1]

- 33 Ethylene glycol, $\text{HOCH}_2\text{CH}_2\text{OH}$, is used as a de-icer. It allows ice to melt at temperatures below 0°C .

Which statements are correct?

- 1 Ethylene glycol disrupts the extensive network of hydrogen bonds in ice.
- 2 Ethylene glycol molecules form hydrogen bonds with other ethylene glycol molecules.
- 3 Ethylene glycol molecules will dissolve in the water formed from the ice.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

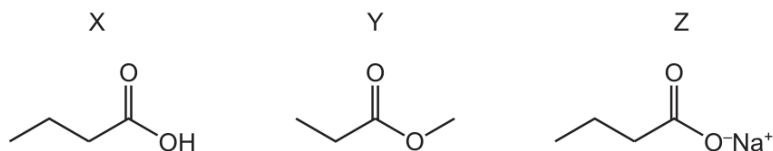
Use of the Data Booklet may be appropriate for some questions.

2021 ON Paper 11 Question 33

A7.

[1]

- 3 The structures represent three compounds, each with four carbon atoms per molecule.



Which row is correct?

	lowest boiling point	→	highest boiling point
A	X	Y	Z
B	Y	X	Z
C	Z	X	Y
D	Z	Y	X

2021 MJ Paper 12 Question 3

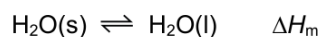
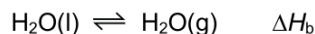
A8.

[1]

- 10** The average intermolecular forces in water are much stronger than the average intermolecular forces in steam.

The average intermolecular forces in ice are slightly stronger than the average intermolecular forces in water.

Enthalpy changes are associated with the equilibrium processes shown.



Which statements are correct?

- 1 ΔH_{b} and ΔH_{m} are both negative.
- 2 ΔH_{b} is greater than ΔH_{m} .
- 3 The intermolecular forces in ice and water are of the same type.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

2025 MJ Paper 12 Question 10

Section B — Structured questions (10 marks)

B1.

[1]

Name the strongest intermolecular force that exists between $\text{NH}_3(\text{l})$ molecules.

2023 MJ Paper 22 Question 1(b)(iii)

B2.

[2]

Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of $\text{NH}_3(\text{l})$. Include any relevant lone pairs of electrons and dipoles.

2023 MJ Paper 22 Question 1(b)(iv)

B3.

[2]

Nitrogen, N_2 , is a gas at room temperature and pressure. Neither CO nor N_2 is an ideal gas.

Complete the table by naming **all** the types of intermolecular forces (van der Waals') in separate samples of $\text{N}_2(\text{g})$ and $\text{CO}(\text{g})$.

	$\text{N}_2(\text{g})$	$\text{CO}(\text{g})$
number of electrons per molecule	14	14
presence of a dipole moment	×	✓
boiling point / °C	−195.8	−191.5
intermolecular forces (van der Waals')		

2021 MJ Paper 22 Question 2(c)(iii)

B4.

[3]

Table 1.1 shows melting points of some oxides.

Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

- i.d. = instantaneous dipole–induced dipole
- p.d. = permanent dipole–permanent dipole

- H = hydrogen bond
- C = covalent bond
- I = ionic bond

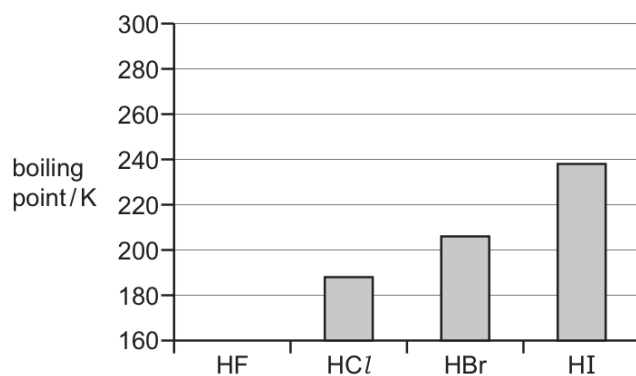
oxide	melting point / °C	force of attraction broken during melting
SO ₂	-73	
H ₂ O	0	
SiO ₂	1610	
Na ₂ O	1132	
MgO	2852	

2025 MJ Paper 22 Question 1(d)(i)

B5.

[2]

- (ii) The bar chart shows the boiling points of HCl, HBr and HI. The boiling point of HF is not shown.



Hydrogen bonds form between HF molecules.

Draw a bar on the bar chart to predict the boiling point of HF.

Explain your answer.

.....
 [2]

2021 ON Paper 22 Question 1(a)(ii)